

EXPENSE TRACKER 2.0 APPLICATION

PYTHON PROGRAMMING INTERNSHIP – TASK 3

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Internship: MAINCRAFTS TECHNOLOGIES

Domain: Python Programming

Task: Task - 3

Project: Expense Tracker 2.0 with Categories & Monthly Reports

Project Overview

This project enhances the Expense Tracker application by adding categories, category-wise search, total spending per category and monthly spending analysis using CSV file handling in Python.

Objectives of Task-3

- Learn CSV file handling and data filtering.
- Implement category-based expense management.
- Generate category-wise and monthly reports.
- Build a menu-driven command line application.
- Improve Python programming skills.

Features Implemented

- Add expenses with categories.
- View all stored expenses.
- Search expenses by category.
- Calculate total spent per category.
- Calculate monthly spending.
- Store records permanently in CSV.

Technologies Used

- Python
- CSV Module
- Datetime Module
- File Handling
- Functions and Loops
- CLI

Task-3 Python Source Code

```
import csv
import os
from datetime import

datetime FILE_NAME =

"expenses.csv"

def initialize_file():
    if not os.path.exists(FILE_NAME):
```

```

        with open(FILE_NAME, "w", newline="") as file:
            writer = csv.writer(file)
            writer.writerow(["Description", "Amount", "Category", "Date"])

def add_expense():
    desc = input("Enter Description: ")
    amount = float(input("Enter Amount: "))
    category = input("Enter Category (Food/Travel/Shopping/etc): ")
    with open(FILE_NAME, "a", newline="") as file:
        writer = csv.writer(file)
        writer.writerow([desc, amount, category, datetime.now().strftime("%Y-%m-%d")])
    print("Expense Added Successfully!")

def view_expenses():
    with open(FILE_NAME, "r") as file:
        reader = csv.reader(file)
        print("\n--- All Expenses ---")
        for row in reader:
            print(row)

def search_category():
    category = input("Enter Category to Search: ")
    with open(FILE_NAME, "r") as file:
        reader = csv.reader(file)
        print(f"\nExpenses in Category: {category}")
        found = False
        for row in reader:
            if len(row) > 2 and row[2].lower() == category.lower():
                print(row)
                found = True
        if not found:
            print("No expenses found.")

def total_per_category():
    totals = {}
    with open(FILE_NAME, "r") as file:
        reader = csv.reader(file)
        next(reader)
        for row in reader:
            category = row[2]
            amount = float(row[1])
            totals[category] = totals.get(category, 0) + amount
    print("\n--- Total Spent Per Category ---")
    for category, amount in totals.items():
        print(f"{category}: ■{amount}")

def monthly_spending():
    month = input("Enter Month (YYYY-MM): ")
    total = 0
    with open(FILE_NAME, "r") as file:
        reader = csv.reader(file)
        next(reader)
        for row in reader:
            if row[3].startswith(month):
                total += float(row[1])
    print(f"Total Spending for {month}: ■{total}")

def menu():

```

```

initialize_file()
while True:
    print("\n===== Expense Tracker 2.0 =====")
    print("1. Add Expense")
    print("2. View Expenses")
    print("3. Search by Category")
    print("4. Total Per Category")
    print("5. Monthly Spending")
    print("6. Exit")
    choice = input("Enter Choice: ")
    if choice == "1":
        add_expense()
    elif choice == "2":
        view_expenses()
    elif choice == "3":
        search_category()
    elif choice == "4":
        total_per_category()
    elif choice == "5":
        monthly_spending()
    elif choice == "6":
        print("Thank You!")
        break
    else:
        print("Invalid Choice!")

menu()

```

Sample Output

```
===== Expense Tracker 2.0 =====
```

1. Add Expense
2. View Expenses
3. Search by Category
4. Total Per Category
5. Monthly Spending
6. Exit

Enter Choice: 1

Enter Description: Food

Enter Amount: 200

Enter Category: Food

Expense Added Successfully!

Enter Choice: 4

--- Total Spent Per Category ---

Food: ■200

Enter Choice: 5

Enter Month (YYYY-MM): 2026-06

Total Spending for 2026-06: ■200

Conclusion

The Expense Tracker 2.0 application successfully manages expenses using categories, searching, category-wise totals and monthly reports. This project improved practical knowledge of Python file handling and data processing.